

01 / AUTONOMOUS ROBOTICS

Ultrasonic Obstacle Avoider Robot

| Non-Contact Distance Sensing & Spatial Avoidance with ESP32.

ACTUATOR 26

FIELD NOTES / 2026



02 / SYSTEM ARCHITECTURE

4 Core Modules of an Obstacle Avider

Real-time distance measurement drives autonomous path clearing:

BRAIN

ESP32 Microcontroller

Triggers 40kHz sonic pulses, measures microsecond echo pulse widths, and computes distance in centimeters.

PERCEPTION

Dual HC-SR04 Sensors

Left and Right ultrasonic modules provide 2D spatial awareness up to 400cm with 3mm precision.

ACTUATION

TB6612FNG Driver

Dual H-Bridge MOSFET driver controlling left and right DC motors to pivot away from obstacles.

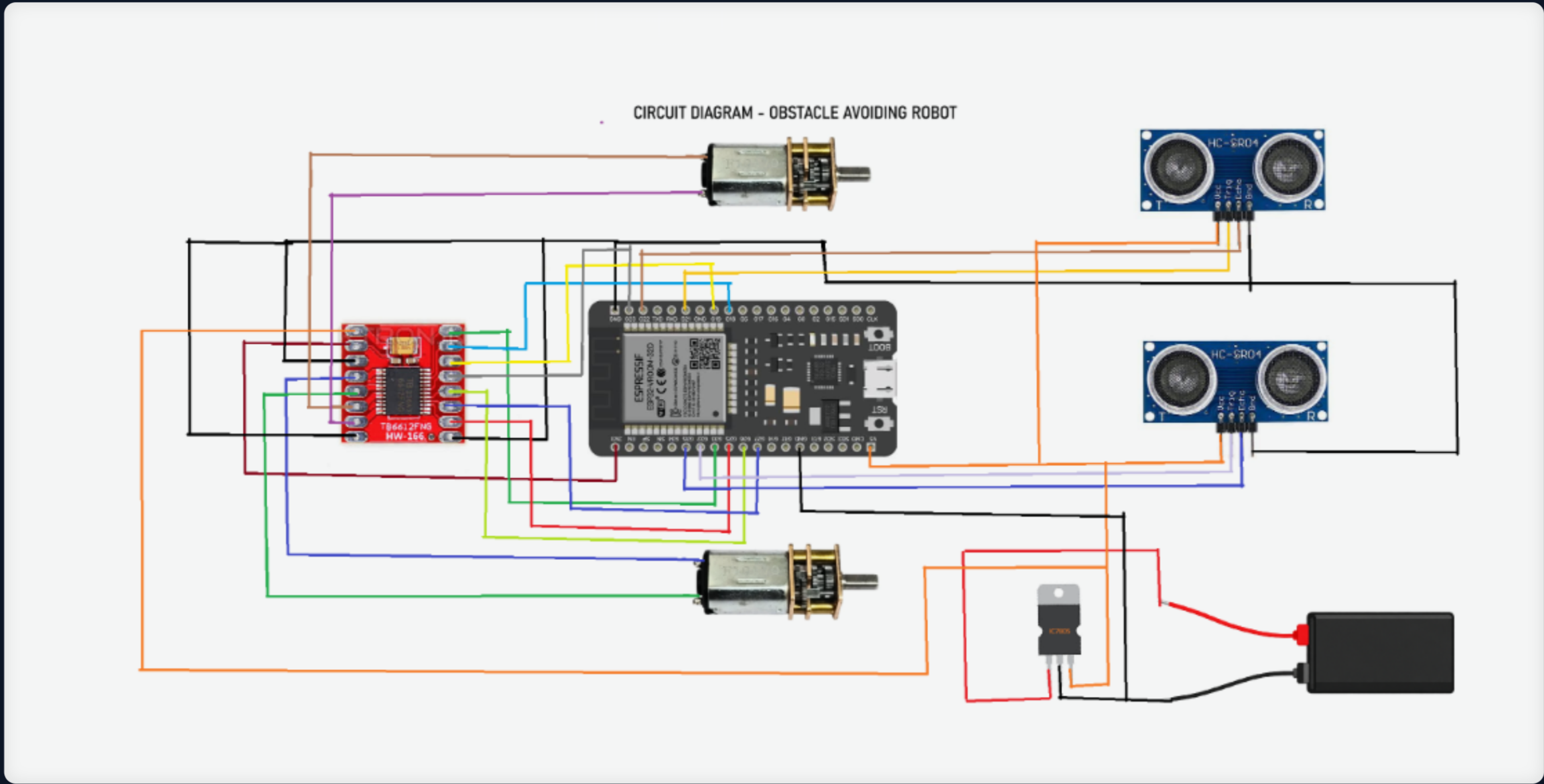
POWER

Isolated LiPo Rail

Delivers clean logic power to ESP32 and high-current motor supply (VM) to avoid brownouts.

03 / SCHEMATIC DIAGRAM

Circuit Diagram & Hardware Interconnects



Left Ultrasonic

Trig: 32 | Echo: 35

Right Ultrasonic

Trig: 21 | Echo: 22

TB6612FNG Driver

STBY: 23 | PWM: 33/25

Threshold Distance

20.0 cm

04 / SENSOR PRINCIPLE

HC-SR04 Ranging Sonic Echo Principle

Computing target distance from sound propagation velocity (343 m/s):

- ↳ **Trigger Pulse**
Send 10µs HIGH pulse on RC_US_TRIG to fire 8-cycle 40kHz sonic burst.
- ↳ **Echo Measurement**
pulseIn(echoPin, HIGH) measures microsecond travel duration.

$$\text{Distance (cm)} = \frac{\text{Duration (}\mu\text{s)} \times 0.0343}{2}$$

// Distance Measurement Helper Function

```
float measureDistance(int trigPin, int echoPin) {  
    digitalWrite(trigPin, LOW);  
    delayMicroseconds(2);  
    digitalWrite(trigPin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trigPin, LOW);  
  
    long duration = pulseIn(echoPin, HIGH);  
    return (duration * 0.0343) / 2; // Convert to cm  
}
```



05 / PIN MAPPING

ESP32 Pinout Assignment

Obstacle Avoider Config

Exact GPIO assignments configured in `obstacle_avoider/src/main.cpp`:

ULTRASONIC SENSORS	
RC_US_LEFT_TRIG	GPIO 32 (Output)
RC_US_LEFT_ECHO	GPIO 35 (Input)
RC_US_RIGHT_TRIG	GPIO 21 (Output)
RC_US_RIGHT_ECHO	GPIO 22 (Input)
THRESHOLD	20.0 cm

MOTOR DRIVER PINS	
RC_MOTOR_STBY	GPIO 23 (Standby Enable)
RC_LEFT_PWM / IN1 / IN2	GPIO 33 / 18 / 19
RC_RIGHT_PWM / IN1 / IN2	GPIO 25 / 26 / 27
SPEED	100 (PWM Duty Cycle)



06 / CONTROL ALGORITHM

Obstacle Avoidance

Spatial Decision Matrix

Dual distance readings drive spatial navigation choices:

LEFT DISTANCE	RIGHT DISTANCE	LEFT MOTOR	RIGHT MOTOR	ROBOT MOVEMENT
> 20 cm (Clear)	> 20 cm (Clear)	FORWARD (SPEED 100)	FORWARD (SPEED 100)	Move Straight Ahead
< 20 cm (Blocked)	> 20 cm (Clear)	FORWARD (SPEED 100)	STOP (0 PWM)	Turn Right (Avoid Left Wall)
> 20 cm (Clear)	< 20 cm (Blocked)	STOP (0 PWM)	FORWARD (SPEED 100)	Turn Left (Avoid Right Wall)
< 20 cm (Blocked)	< 20 cm (Blocked)	STOP (0 PWM)	STOP (0 PWM)	Emergency Stop / Dead End

07 / CODE BREAKDOWN

Initialization & Ultrasonic Setup Code

Configuring Trig/Echo pins & motor driver enable in setup():

```
// setup() from obstacle_avoider/src/main.cpp

void setup() {
  Serial.begin(115200);
  pinMode(RC_MOTOR_STBY, OUTPUT);
  pinMode(RC_LEFT_PWM, OUTPUT);   pinMode(RC_LEFT_IN1, OUTPUT);   pinMode(RC_LEFT_IN2, OUTPUT);
  pinMode(RC_RIGHT_PWM, OUTPUT);  pinMode(RC_RIGHT_IN1, OUTPUT);  pinMode(RC_RIGHT_IN2, OUTPUT);

  pinMode(RC_US_LEFT_TRIG, OUTPUT); pinMode(RC_US_LEFT_ECHO, INPUT);
  pinMode(RC_US_RIGHT_TRIG, OUTPUT); pinMode(RC_US_RIGHT_ECHO, INPUT);

  digitalWrite(RC_MOTOR_STBY, HIGH); // Enable TB6612FNG Driver
}
```

08 / CODE BREAKDOWN

Dual Distance Sampling & Serial Telemetry

Sampling dual ultrasonic sensors in loop():

```
// Sampling Dual Distances in loop()

void loop() {
  float leftDistance = measureDistance(RC_US_LEFT_TRIG, RC_US_LEFT_ECHO);
  float rightDistance = measureDistance(RC_US_RIGHT_TRIG, RC_US_RIGHT_ECHO);

  Serial.print("Left: "); Serial.print(leftDistance);
  Serial.print(" | Right: "); Serial.println(rightDistance);

  // Spatial Avoidance Logic Follows...
}
```

09 / CODE BREAKDOWN

Spatial Avoidance Decision Tree Code

Executing directional avoidance based on distance thresholds:

```
if (leftDistance < THRESHOLD_DISTANCE && rightDistance < THRESHOLD_DISTANCE) {  
    // Both blocked -> Stop  
    digitalWrite(RC_LEFT_IN1, LOW); digitalWrite(RC_LEFT_IN2, LOW); analogWrite(RC_LEFT_PW  
    digitalWrite(RC_RIGHT_IN1, LOW); digitalWrite(RC_RIGHT_IN2, LOW); analogWrite(RC_RIGHT_P  
} else if (leftDistance < THRESHOLD_DISTANCE) {  
    // Left blocked -> Turn Right  
    digitalWrite(RC_LEFT_IN1, HIGH); digitalWrite(RC_LEFT_IN2, LOW); analogWrite(RC_LEFT_PW  
    digitalWrite(RC_RIGHT_IN1, LOW); digitalWrite(RC_RIGHT_IN2, LOW); analogWrite(RC_RIGHT  
} else if (rightDistance < THRESHOLD_DISTANCE) {  
    // Right blocked -> Turn Left  
    digitalWrite(RC_LEFT_IN1, LOW); digitalWrite(RC_LEFT_IN2, LOW); analogWrite(RC_LEFT_P  
    digitalWrite(RC_RIGHT_IN1, HIGH); digitalWrite(RC_RIGHT_IN2, LOW); analogWrite(RC_RIGHT_  
} else {  
    // Clear -> Move Forward  
    digitalWrite(RC_LEFT_IN1, HIGH); digitalWrite(RC_LEFT_IN2, LOW); analogWrite(RC_LEFT_P  
    digitalWrite(RC_RIGHT_IN1, HIGH); digitalWrite(RC_RIGHT_IN2, LOW); analogWrite(RC_RIGHT_  
}  
delay(100);
```

10 / HARDWARE TUNING

Threshold Tuning & Sensor Calibration

DISTANCE THRESHOLD**20.0 cm Threshold**

Provides optimal braking distance at 100 PWM motor speed. Increase threshold for faster speeds.

SENSOR ANGLE**15° Toe-Out Angle**

Slightly angling left and right sensors outward expands forward field-of-view for corner detection.

TIMEOUT PROTECTION**pulseIn Timeout**

If no reflection returns within 30ms, pulseIn returns 0 to prevent code freezing.

GPIO 35 INPUT ONLY**ESP32 Pin Constraint**

GPIO 35 is an input-only pin without internal pull-ups. Ideal for HC-SR04 Echo output signal.

11 / TROUBLESHOOTING

4 Ultrasonic Field Diagnostics

ISSUE 1**Soft Surface Absorption**

Curtains or foam absorb 40kHz sound waves. Test obstacle avoidance against rigid surfaces.

ISSUE 2**Crosstalk & Interference**

Triggering left and right sensors simultaneously can cause acoustic crosstalk. Space measurements out.

ISSUE 3**5V to 3.3V Echo Level Shift**

HC-SR04 Echo pin outputs 5V. Use a resistor divider (1k/2k) when connecting Echo to ESP32 3.3V GPIOs.

ISSUE 4**Power Supply Noise**

Motor electrical noise on VM rail can disrupt ultrasonic pulse timing. Add 100μF decoupling capacitor across VM/GND.



12 / READY TO AVOID

Sense. Decide. Avoid.

Your ESP32 Ultrasonic Obstacle Avider is configured for spatial 2D navigation.

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LET'S
AVOID IT